3.1 How sensors work

Modern sensors enable digital twins to monitor various aspects of physical systems. Building management requires environmental monitoring (temperature, humidity, occupancy) to optimize heating, ventilation and air-conditioning (HVAC) systems. Equipment health monitoring relies on motion and vibration sensors (accelerometers, gyroscopes) to detect anomalies before failure. Environmental quality monitoring uses chemical sensors to track air pollutants and water conditions. Energy management depends on electrical sensors measuring current, voltage, and power consumption. Spatial monitoring employs optical sensors (cameras, light detectors) and ultrasonic sensors for occupancy and distance measurement. Effective digital twins require strategic sensor selection, deploying only sensors that provide data necessary for the specific decisions and actions the system must take.

What distinguishes a digital twin from traditional monitoring systems is both purpose and scale. Traditional monitoring systems passively track metrics and alert when thresholds are breached. Digital twins go further by enabling active control based on continuous feedback. They require:

- *Data density*—comprehensive streams that enable predictive capabilities through machine learning.
- *Simulation capabilities*—rich historical data for scenario modeling and what-if analysis.
- *Timely synchronization*—frequent updates that keep the digital model current with the physical system.
- *Contextual understanding*—multiple data streams that reveal relationships between system components.

No matter which type of sensor you choose, they all follow a similar path from physical measurement to digital data, as shown in figure 3.1.